

terms (*chelandion*, *naus*, *karabion*, *ploion*, *trieres*, *monoxylon*, *galea*, etc.) when they wrote of Byzantine fleets and naval actions?

The complexity of these questions is evident from the scope of the inquiry. The authors begin with a review of the contest between the Byzantines and a succession of rivals (Germanic tribes, Muslim invaders, crusaders, and the Italian maritime powers) for control of the Mediterranean. They then index the evidence for the origins of the *dromon*, a mono-rème warship distinguished from its predecessors by features confirmed by sixth-century sources, namely, the replacement of the square sail by the triangular lateen sail; a full deck covering the oarsmen below; and changes in the design of the hull, most noticeable at the bow where the ram gave way to an above-water spur that enabled the *dromon* to ride up over the oars of an enemy vessel. The evidence is too scanty to trace the development of the *dromon* between the seventh and ninth centuries, but the relative abundance of source material from the tenth century allows the authors to dissect the principal Byzantine warship from bow to stern. The discussion of each of the ship's parts expands into a broader consideration of Byzantine ship construction, seafaring, and battle tactics, which makes the book extraordinarily valuable to specialists and nonspecialists alike. With remarkable expertise and a relentlessly commonsensical approach, the authors present and evaluate the often problematic sources, which, when carefully sifted, portray the *dromon* in its tenth-century version as a two-masted, bireme galley equipped with an impressive panoply that included the fearsome Greek fire. The speed and range of the *dromon*, the effort and skills required to sail it, the provisioning of its crew, and, above all, the risks and factors governing the conduct of amphibious campaigns are among the many practical aspects of Byzantine maritime history that the authors illuminate for the first time.

The authors have made every effort to make their work accessible to readers of many backgrounds. Greek terms are transliterated (except, oddly enough, in the title), and a full glossary explains the intimidating lexicon of nautical terms and functions. Copious illustrations and drawings aid comprehension of technical details. Equally valuable are the appendices containing the texts and annotated translations of the treatises on naval warfare, which have until now been the least studied of the Byzantine military writings. An Arabic naval treatise based on the *Tactica* of Leo VI complements the Greek sources, and the authors have reserved appendix 6 for a thorough appraisal of Greek fire and the means by which it was employed in naval engagements.

It is difficult to overstate the importance of this book. The authors have scoured the literary, artistic, and archaeological sources, clarified much obscure terminology, and clearly laid out the evidence on which they base their findings. Byzantinists now have a reliable guide to the technical, operational, and strategic dimensions of the empire's navy, while scholars of the medieval West and Islam will find much helpful material for the study not only of the military but of the cultural and economic relations between Byzantium and its neighbors. John Pryor and Elizabeth Jeffreys deserve high praise for putting such a wealth of material at the disposal of all scholars interested in the medieval Mediterranean.

ERIC MCGEER, St. Clement's School, Toronto

MARCO RAININI, *Disegni dei tempi: Il "Liber figurarum" e la teologia figurativa di Gioacchino da Fiore*. (Opere di Gioacchino da Fiore: Testi e Strumenti, 18.) Rome: Viella, 2006. Paper. Pp. xv, 333 plus black-and-white and color figures; tables. €35.

In 2004 Gian Luca Potestà published an important book on the Calabrian apocalyptic thinker Joachim of Fiore in which he traced Joachim's intellectual pilgrimage, making effective use of his minor works to reveal the layers of development encased in his three major ones. In the present work Marco Rainini applies the same technique to a series of

figurae Joachim employed to present his thought in visual form. Most of us know these *figurae* as Marjorie Reeves and Beatrice Hirsch-Reich described them close to a half century ago, and as a result we are inclined to think many of them were combined into a single work produced late in Joachim's life, the *Liber figurarum*. The fact that Rainini places the words *Liber figurarum* in quotation marks goes a long way toward suggesting what he thinks of that view. He agrees that Joachim produced individual *figurae* but thinks it likely that they were collected together only after Joachim's death, and he sees the collection as more like a portfolio of designs or envelope of photographs than like a book with its own internal consistency.

Following in Potestà's wake, Rainini uses the minor works to date individual figures and in the process shows how both the works and the figures illustrate various moments throughout the long, slow evolution of Joachim's thought. Here the story becomes complicated. Like the figures, the minor works can be seen as what Rainini calls "container texts" in which Joachim or his disciples (Rainini favors the latter view) combined heterogeneous materials written at different times. Thus the problem is to compile a chronology not only of individual works but of parts of these works. At any rate, of the *figurae* he considers, he dates the earliest around 1176—which is remarkably early—and the latest in the late 1190s. Joachim died in 1202.

Rainini does two things very well: dating the figures and explaining what they mean. In those respects his work is a valuable extension of Potestà's. Along the way he does other things competently though more briefly. For example, he begins with a survey of recent scholarship on what he and others call "visual exegesis," the use of *figurae* to present complex interpretations of Scripture and, more generally, of Christian doctrine. In the course of the work he at least makes a start on explaining how such figures were used: for didactic purposes, but also for individual meditation. He also has some thoughts on the question of which authors Joachim was consulting—in fact, he makes interesting comparisons with the Victorines, especially Hugh—and he does a good job of explaining how Joachim would have had access to good libraries. The Joachim who emerges from this book is a man who, although an autodidact in the sense that he was a mature adult when he entered monastic life and thus was not shaped by the standard monastic education, nonetheless had the resources to do a great deal of catching up on his own.

Little of this is set in stone. Rainini's major project of dating and interpretation will undoubtedly be challenged in some of its particulars because this sort of thing is very hard to do, and someone will almost certainly find room for complaint. That does not detract from Rainini's achievement, however.

DAVID BURR, Virginia Polytechnic Institute and State University

JONATHAN RAY, *The Sephardic Frontier: The "Reconquista" and the Jewish Community in Medieval Iberia*. (Conjunctions of Religion and Power in the Medieval Past.) Ithaca, N.Y., and London: Cornell University Press, 2006. Pp. xiii, 198 plus black-and-white frontispiece; maps. \$35.

In this impeccably researched and clearly expressed study of Jewish communities on the Aragonese, Castilian, Portuguese, and Valencian frontiers, Jonathan Ray has adroitly traversed the historiographical chasms left by Américo Castro and David Nirenberg and safely found his way to the other side. By clearly focusing on the documentary record and eschewing the temptation to assign an exceptional record of achievement or victimization to the communities he studies, Ray clearly places Sephardic Jewry in an Iberian frontier context written of so eloquently by Robert I. Burns, S.J., Archibald Lewis, Derek Lomax, and Angus MacKay. By so doing, he closely examines all sides of the issues of *Reconquista* and